

# Factorization into $A = LU$

Record elimination **once**, then solve for any new target in two quick passes.

Shown on a real money question: the same funds, many clients.

L from the multipliers

Factor once, solve many

MIT 18.06, Lecture 4

## THE PROBLEM

# Same funds, new targets every day

Nifty, bond and midcap funds; rules for budget, return and market exposure. The matrix  $A$  never changes:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}$$

But each client brings a different target  $b$ : ₹10 lakh at 11.1%, ₹10 lakh at 9.8%, ₹15 lakh...

### THE IDEA

Don't redo elimination for every client. Save it as  $A = LU$ .

# Undo in reverse order

$$(AB)^{-1} = B^{-1}A^{-1}$$

Check:  $AB B^{-1}A^{-1} = A I A^{-1} = I$ .

## SHOES AND SOCKS

Socks on, then shoes. To undo: shoes off first, then socks.

Elimination applied  $E_{21}, E_{31}, E_{32}$ . To undo it:  $E_{32}^{-1}$  first,  $E_{21}^{-1}$  last.

2 × 2

## From $EA = U$ to $A = LU$

$$\begin{bmatrix} 1 & 0 \\ -12 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}$$

Move  $E$  to the other side (its inverse adds back 12 × row 1):

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}}_U$$

Lower triangular with the multiplier; Upper triangular with the pivots.

3 × 3

# The multipliers go straight into **L**

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \mathbf{12} & 1 & 0 \\ \mathbf{1} & \mathbf{0.2} & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}}_U$$

## KEY FACT

No row exchanges  $\Rightarrow$  the multipliers  $\ell_{21} = 12$ ,  $\ell_{31} = 1$ ,  $\ell_{32} = 0.2$  sit directly in  $L$ . No extra work.

## E is messy, L is clean

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 1.4 & -0.2 & 1 \end{bmatrix} \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 1 & 0.2 & 1 \end{bmatrix}$$

Going forward, steps interfere: removing  $0.2 \times$  the new row 2 quietly adds back  $0.2 \times 12 = 2.4$  of row 1, so  $-1 + 2.4 = 1.4$ .

Going backward with  $L$ , nothing interferes. That's why software stores  $L$  and  $U$ .

# Two triangular passes

Forward:  $Lc = b$ , top down. Back:  $Ux = c$ , bottom up.

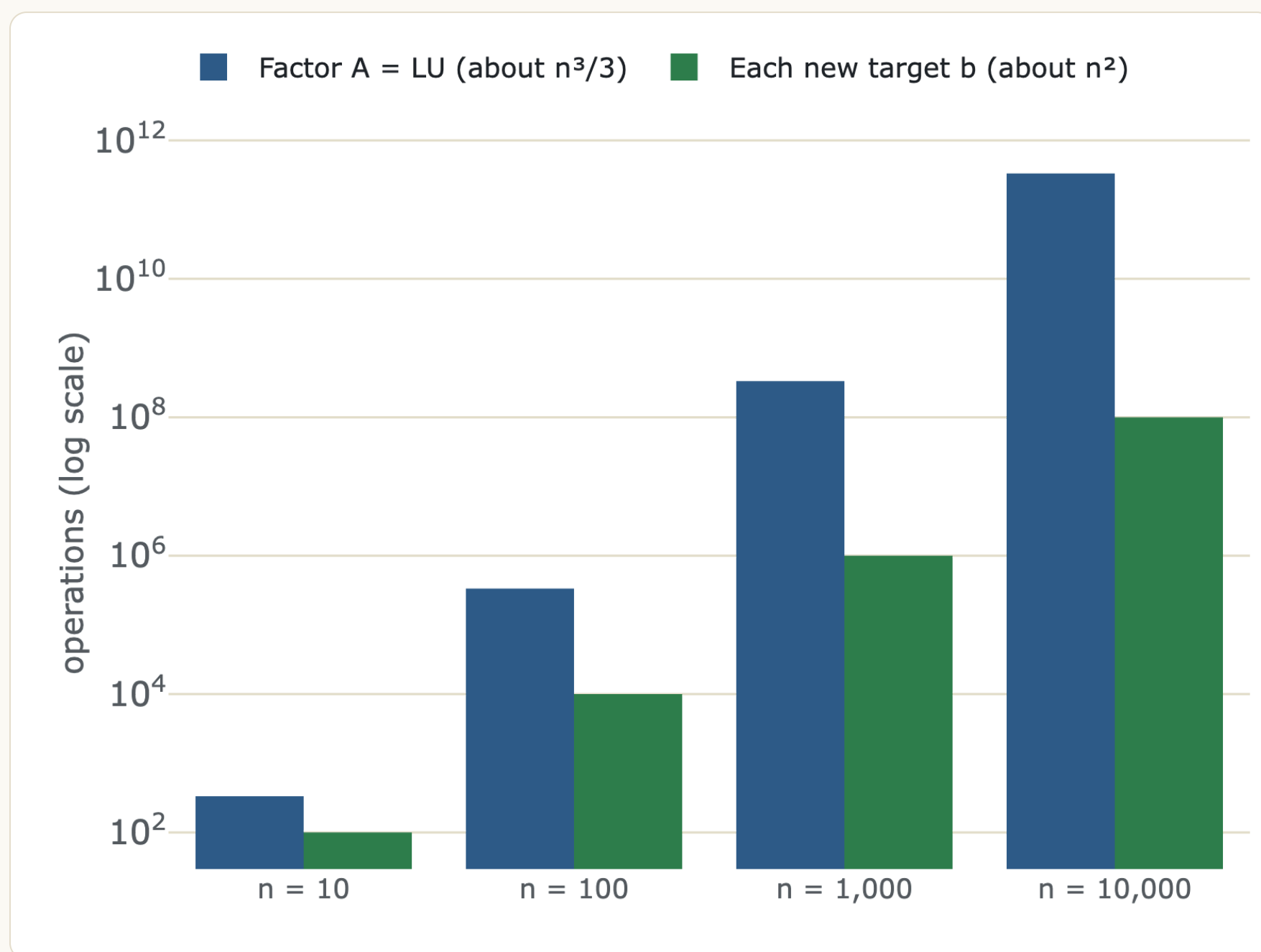
Client:  $b = (10, 98, 5.2)$

- $c_1 = 10, c_2 = 98 - 120 = -22, c_3 = 5.2 - 10 + 4.4 = -0.4$
- $z = 1$ , then  $-5y + 3 = -22 \Rightarrow y = 5$ , then  $x = 4$

**₹4 lakh Nifty, ₹5 lakh bonds, ₹1 lakh midcap.** No new elimination needed.



# Factor once, solve many



Factoring: about  $\frac{1}{3}n^3$  operations. Each new target: about  $n^2$ . For 1,000 assets: 333 million once, then 1 million per client.



# **$A = LDU$**

Pull the pivots out of  $U$  into a diagonal matrix  $D$ :

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -0.4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -0.6 \\ 0 & 0 & 1 \end{bmatrix}$$

Now both triangular factors have 1s on the diagonal, and the pivots sit in the middle.

## ROW EXCHANGES

$$PA = LU$$

A credit fund (12%, like Nifty) puts a 0 in a pivot position. Swap rows first, then factor:

$$P_{23}A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 12 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix}$$

### PERMUTATIONS

$n!$  of them (6 for  $3 \times 3$ ). Closed under multiplication.

$$P^{-1} = P^T.$$

WATCH OUT

# Singular matrices can have LU too

The balanced fund (half Nifty, half bonds) factors with the same  $L$ :

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$

The trouble shows up on  $U$ 's diagonal: a 0 pivot, so back substitution fails.

## TEST

All pivots on  $U$ 's diagonal non-zero  $\Leftrightarrow$  invertible.

# Key takeaways

1.  $(AB)^{-1} = B^{-1}A^{-1}$ : reverse order.
2.  $A = LU$ : multipliers in  $L$ , pivots in  $U$ .
3.  $L$  is clean;  $E$  is messy.
4. Solve: forward  $Lc = b$ , back  $Ux = c$ .
5.  $n^3/3$  to factor,  $n^2$  per target.
6. Row swaps:  $PA = LU$ .

QUANT FROM FIRST PRINCIPLES

# The full interactive lesson is free

A lab that factors your own matrix into  $L$  and  $U$  and solves for any target, plus worked hedging and options examples:

**[learn.krisarj.com](https://learn.krisarj.com)**

Based on MIT OpenCourseWare 18.06 Linear Algebra (Prof. Gilbert Strang), used under CC BY-NC-SA 4.0; these slides are shared under the same licence. Not affiliated with or endorsed by MIT. Educational only, not investment advice.